Twitter/X Infrastructure Collapse Under Musk (2022–2023)

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Introduction

When Elon Musk completed his $44 billion acquisition of Twitter in October 2022, he moved quickly and dramatically. Within weeks, the company had laid off roughly half of its workforce. Within months, thousands more had either been let go or had resigned in protest. What followed was a fascinating and troubling real-world experiment: what happens to a complex, global technology platform when you remove a significant portion of the people who build and maintain it?

This case is not primarily about one person's management style. It is about the invisible, underappreciated role that skilled technical workers play in keeping critical systems alive — and what happens to those systems when that knowledge walks out the door.

Background: What Is (Was) Twitter?

Before the acquisition, Twitter was one of the world's most prominent social media platforms, with approximately **230 million daily active users**. It was considered essential infrastructure for public discourse — used by journalists, politicians, businesses, celebrities, academics, and ordinary citizens to communicate in real time.

Behind the scenes, Twitter ran on an enormously complex technology infrastructure. The platform processed hundreds of millions of tweets per day, managed petabytes of data (one petabyte equals one million gigabytes), and had to remain available 24 hours a day, 7 days a week, across every country in the world.

Keeping that running required thousands of specialized engineers — people with deep, specific knowledge of Twitter's unique systems, code, and architecture. Much of this knowledge existed not in documents but **in the heads of the people who built it.**

The Acquisition and the Layoffs

Musk took over Twitter on October 27, 2022. Almost immediately, he moved to dramatically cut costs. His stated goal was to make the company profitable — Twitter had been losing money for years. Within the first week:

- The entire **board of directors** was dissolved
- The **CEO, CFO, and several other top executives** were fired
- Roughly **3,700 employees — about half the workforce** — were laid off in a single wave

The layoffs were chaotic. Many employees learned they had been let go when their work laptops were remotely disabled and their access to company systems was revoked. Some were accidentally laid off and then asked to return. Teams responsible for critical infrastructure were cut without apparent understanding of what those teams actually did.

Shortly after the mass layoffs, Musk issued an ultimatum to remaining employees: **sign a pledge to work "extremely hardcore" with long hours and high intensity, or leave with severance.** Hundreds chose to leave. This second wave of departures further depleted the engineering teams.

The company also faced the loss of a significant portion of its **trust and safety team** — the group responsible for moderating content, enforcing rules, and managing the platform's relationship with advertisers and regulators.

What Happened to the Platform?

The consequences became visible almost immediately.

Outages and Performance Degradation

Twitter began experiencing **frequent outages** — periods when the service was partially or completely unavailable. In the weeks and months following the acquisition, users encountered:

- Intermittent failures to load tweets or timelines
- Errors when trying to send tweets or direct messages
- The platform becoming completely inaccessible for periods of time
- Slowdowns during high-traffic events (such as major sporting events or breaking news)

Some of these outages were spectacular. During the 2023 FIFA Women's World Cup and other major events, Twitter struggled under the load of users tweeting simultaneously — something the old engineering team had routinely managed.

The "Rate Limiting" Crisis

In July 2023, Twitter abruptly limited how many tweets users could read per day — a measure Musk said was necessary to combat "data scraping" (when automated programs harvest large amounts of data from a platform). Verified accounts could read 6,000 tweets per day; unverified accounts just 600. The sudden restriction made the platform nearly unusable for many users and was widely seen as a sign of deeper technical problems rather than a deliberate policy choice.

Failed Feature Launches

New features introduced after the acquisition were frequently buggy or broke existing functionality. The rebranding of Twitter to **"X"** in July 2023 — a rapid name and logo change — caused widespread confusion and technical issues. The checkmark verification system was overhauled multiple times in chaotic fashion, at one point causing impersonation of major brands and celebrities that led to serious incidents.

Advertiser Exodus

Many major advertisers paused or stopped spending on Twitter/X, citing concerns about content moderation after the trust and safety team was gutted. The loss of advertising revenue compounded the company's financial difficulties.

Why Did This Happen? An MIS Perspective

1. The Hidden Value of Institutional Knowledge

Institutional knowledge refers to the accumulated understanding that employees develop over time about how an organization's systems, processes, and culture work. In technology companies, this knowledge is especially critical — engineers who have spent years working on a system understand its quirks, its fragile points, and the history of decisions that shaped it.

When thousands of Twitter's engineers left in a short period, that knowledge left with them. Systems that had been maintained by people who understood them deeply were now maintained by a fraction of the original team, many of whom were new or unfamiliar with Twitter's specific architecture.

2. Technical Debt

Technical debt is a concept in software development that refers to shortcuts, outdated code, or imperfect solutions that accumulate over time when development is done quickly rather than carefully. Every technology company has some technical debt. Managing it requires ongoing engineering work.

Twitter, like many large platforms, had considerable technical debt built up over its 16-year history. The teams that had been managing and slowly paying down that debt were now gone. Maintenance work stopped. Problems that had been held at bay began to surface.

Think of it like deferred maintenance on a building — if you stop fixing small problems, they become big problems.

3. Workforce Reduction Without Impact Assessment

A fundamental principle of IT management is understanding **dependencies** — which systems and functions depend on which other systems and people. Before making major changes, responsible management asks: "If we remove this, what breaks?"

The Twitter layoffs appear to have been executed with limited understanding of which engineering functions were critical to platform stability. Teams that appeared redundant on an organizational chart were in fact performing essential maintenance and monitoring tasks.

4. Change Management at Scale

Musk's approach to Twitter represented perhaps the most aggressive organizational change management experiment in the history of a major technology company. While change is often necessary, best practice suggests that significant changes should be **planned, communicated, staged, and monitored** — not executed simultaneously across all dimensions at once.

Changing ownership, leadership, workforce size, product strategy, monetization model, and brand identity all at the same time virtually guaranteed instability.

The Broader Lessons

The Twitter/X case is a cautionary tale with implications beyond one company:

- **People are infrastructure.** In a technology company, skilled engineers are as critical as the servers and software they maintain. Losing them rapidly is as dangerous as losing physical hardware.
- **Visible costs vs. hidden costs.** Layoffs reduce the visible cost of salaries. But the hidden costs — lost knowledge, system failures, reduced advertiser confidence, regulatory risk — can exceed the savings.
- **Platform trust is fragile.** Users and advertisers stayed on Twitter for years partly out of habit and partly because alternatives were inconvenient. Persistent instability gave them reason to reconsider.

Key Concepts for Discussion

Concept	Definition	Relevance to This Case
Institutional Knowledge	Accumulated expertise held by employees	Mass departures wiped out critical system knowledge
Technical Debt	Accumulated shortcuts in code that require future work	Left unmanaged after engineers departed
IT Change Management	Structured approach to organizational and system changes	Changes were rapid, simultaneous, and poorly sequenced
System Dependency	The reliance of one system/function on another	Layoffs cut critical functions without understanding dependencies
Business Continuity	Maintaining operations through disruptions	Twitter had no continuity plan for rapid workforce loss

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Analysis Questions

1. Describe the company featured in the case, including its competitive advantage (or "moat"), historical background, and the nature of its industry or business operations.

2. Summarize the key facts presented in the case. Do these facts relate to or mirror any other cases discussed in class? If so, what insights or lessons from those previous cases can be applied here?

3. Analyze the economic consequences of the information system failure described. If a similar failure occurred in today's technological environment, would the impact be greater or smaller? How likely is such an incident to occur now?

4. Discuss the broader implications of the risks identified in the case for contemporary IT infrastructures, cloud computing services, and emerging artificial intelligence systems.

5. Identify the key human decision point in this case — the moment where a different choice by a person or team would most likely have prevented or significantly reduced the harm. Who was responsible, and what should they have done differently?

6. Identify and explain five key lessons or takeaways derived from this case study.

Discussion Questions

1. What does this case reveal about where organizations tend to underestimate risk — in their technology, their processes, or their people? Use specific evidence from the case to support your answer.
2. If you were the CIO of the organization in this case, what are the first three changes you would make after this incident, and why?
3. How would you attempt to quantify the value of institutional knowledge during an acquisition due diligence process? What information would you need, and what methods would you use?
4. Musk argued Twitter had too many employees and needed dramatic cost cuts. Is there a responsible way to rapidly reduce a technology company's workforce without destabilizing its systems? What would that look like in practice?
5. Twitter's outages were visible and embarrassing. What other, less visible consequences might a technology platform face after losing a large portion of its engineering team?
6. Many Twitter employees voluntarily left rather than accept the "extremely hardcore" work pledge. What does this tell us about the relationship between organizational culture and the retention of critical IT talent?

Suggested Further Reading

- "Inside Twitter's Meltdown" — *The New York Times*, 2022–2023
- Overview of IT change management frameworks (ITIL, COBIT)

Instructions

1. Use **Bloomberg News** as the primary source for case research. You may also reference other *reputable* public websites or databases. All sources must be properly cited in **APA format** within your written report.
2. The **written case report** should be between **5 and 10 double-sided pages**, inclusive of tables, figures, infographics, or any analytical materials. Acceptable submission formats include **Canva, Microsoft Word, PDF, Markdown, or LaTeX**.
3. The **case presentation** should consist of **10 to 30 slides**, incorporating tables, charts, infographics, or analytical visuals as needed. Acceptable file formats include **Canva, Microsoft PowerPoint, PDF, Quarto Markdown, or LaTeX**.
4. Include a **detailed log of each team member's contributions** as an appendix to the written report. Every team member must be prepared to address questions during the presentation, both on the overall case content and specifically on the sections they contributed to.
5. **AI tools may not be used** to generate written content for the report. Collaborative work through **Google Docs** (with version control and simultaneous editing) is encouraged to demonstrate authentic team contribution and transparency in authorship. All version control tracking links must be shared with the instructor when asked.
